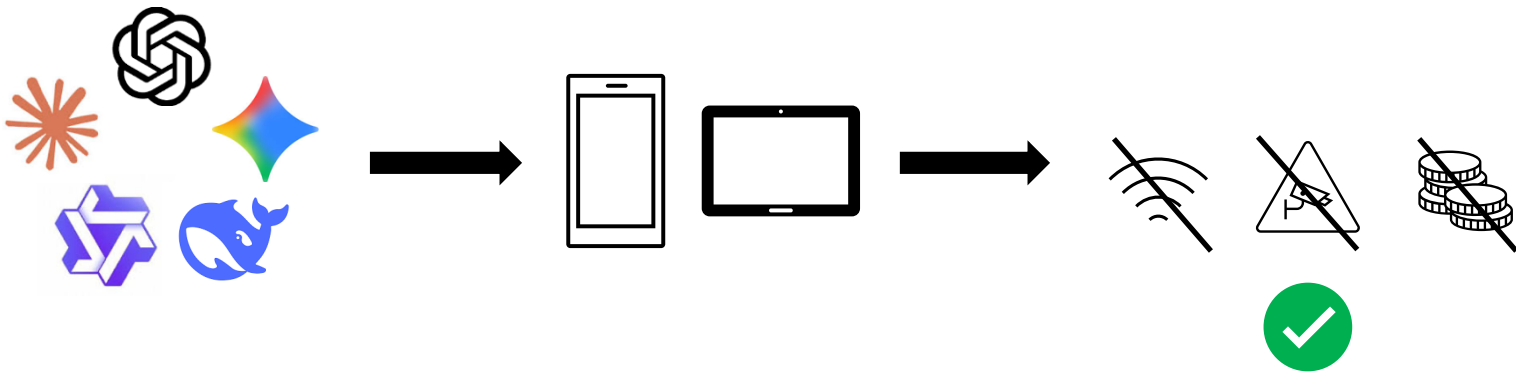


Background

- Large Language Models (LLMs) on edge devices

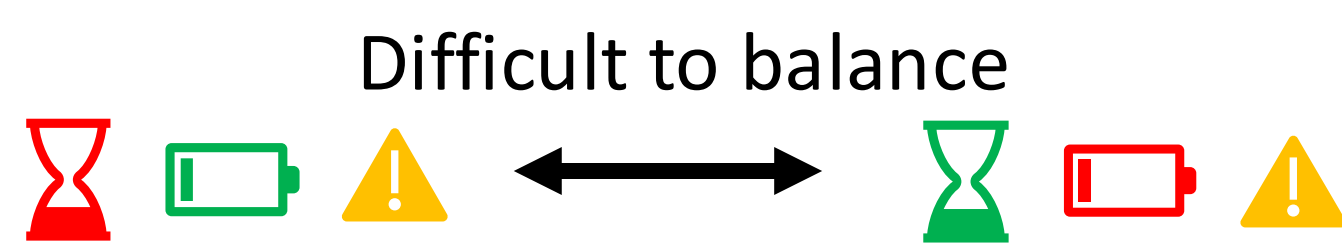


- Challenges:



Compute Battery Overheat

- Limitations for Existing DVFS:

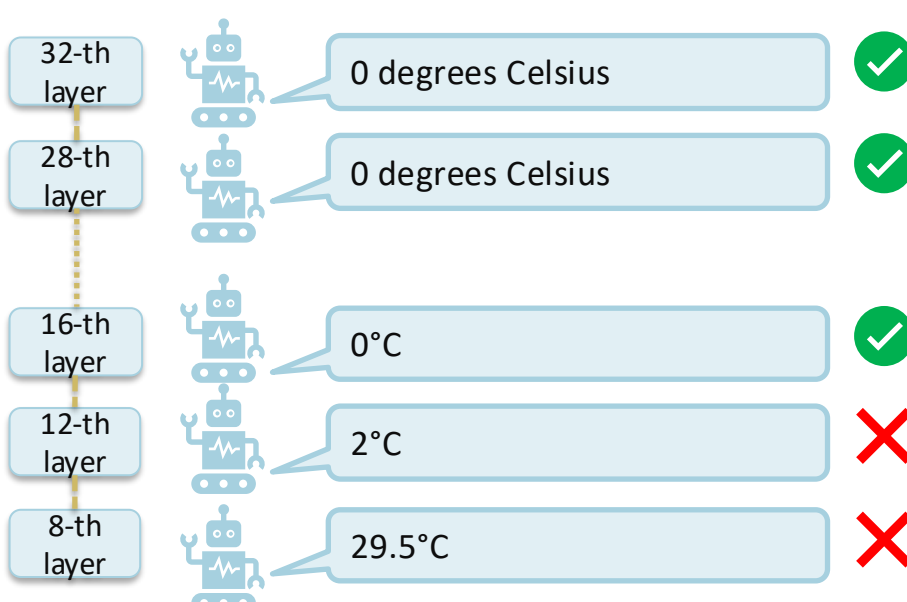


Key Insight and Observations

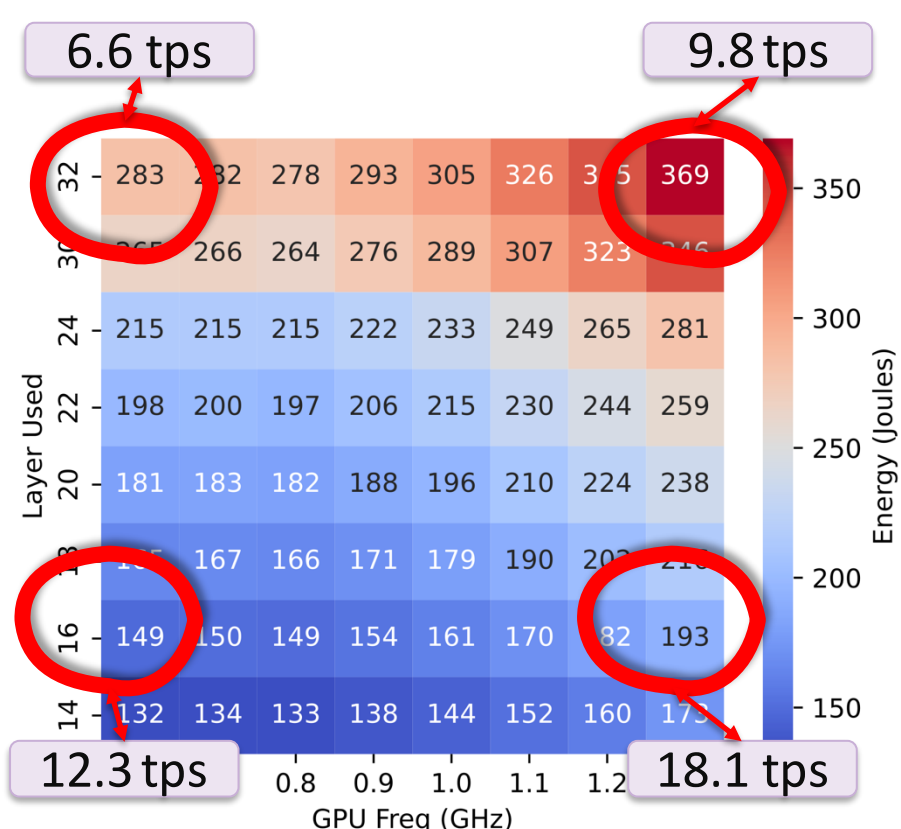
- LLM workload is shapeable:

LLM decoding

Q: What temperature does water freeze at?



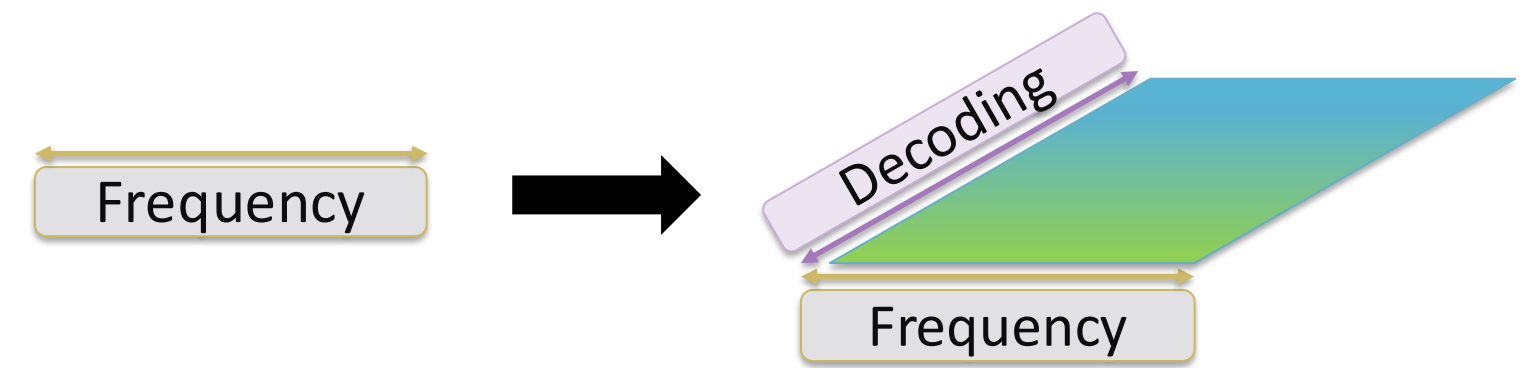
Hardware



(tps: tokens per second)

Our Method

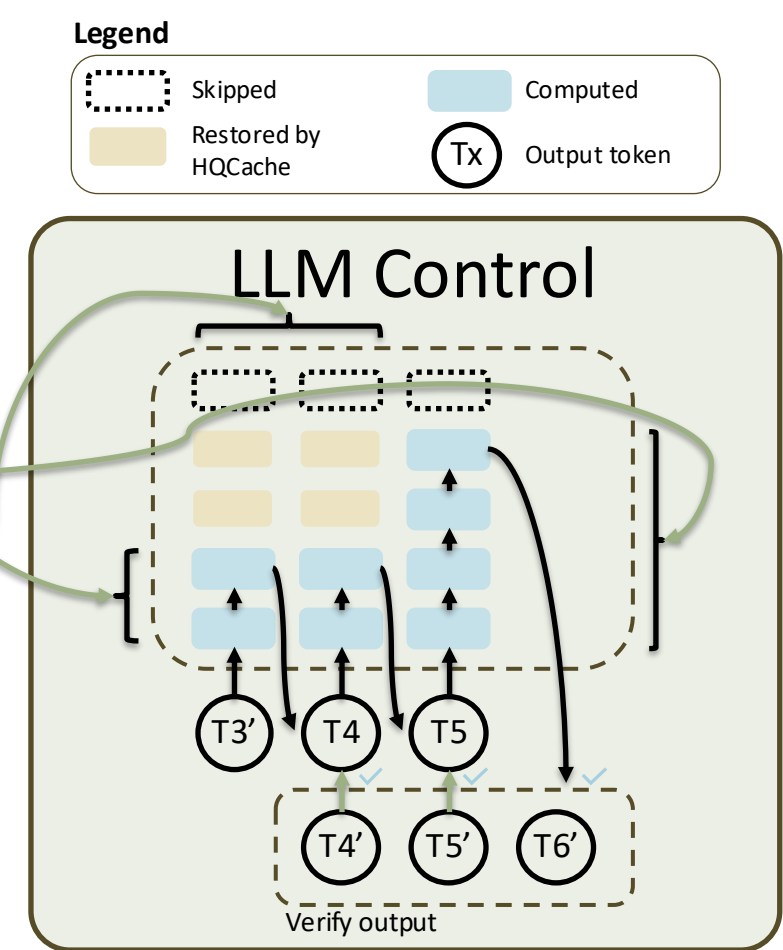
- PELM: Power Efficient Large Language Model Inference Framework for edge devices



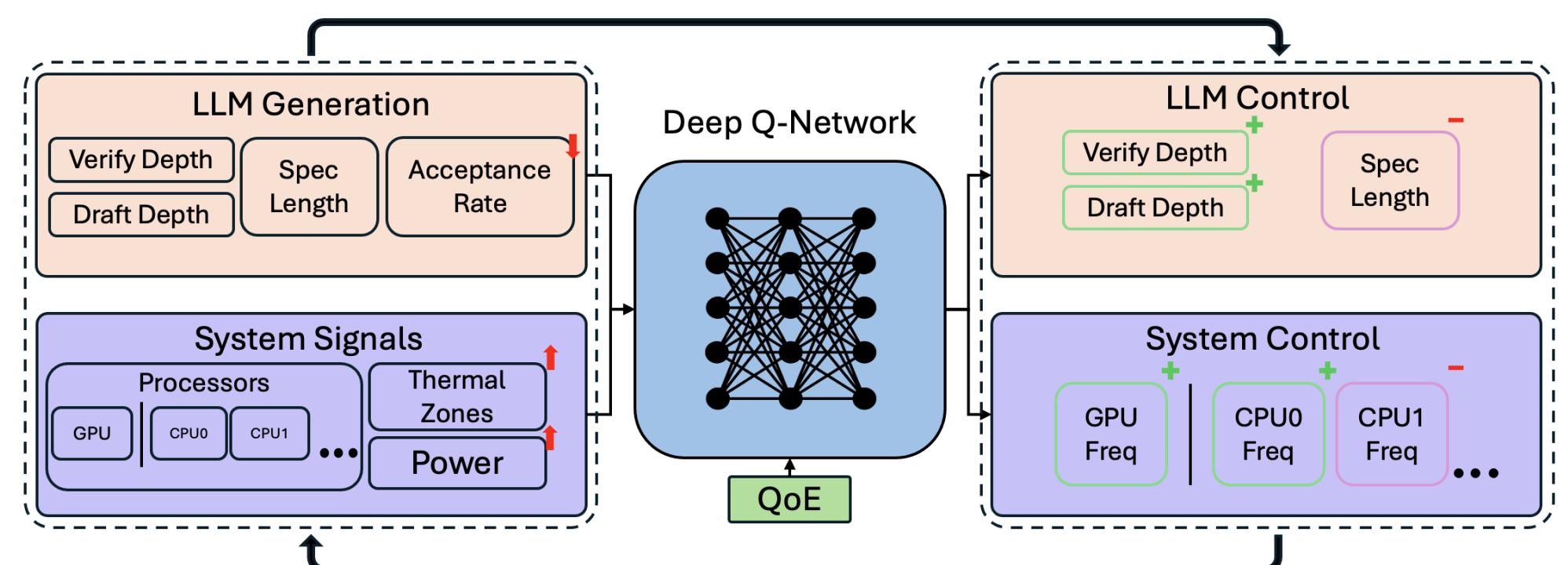
PELM expands the optimization space

- Mechanism 1: Variable-depth Speculative Decoding

- Draft Depth
- Verify Depth
- Spec Length



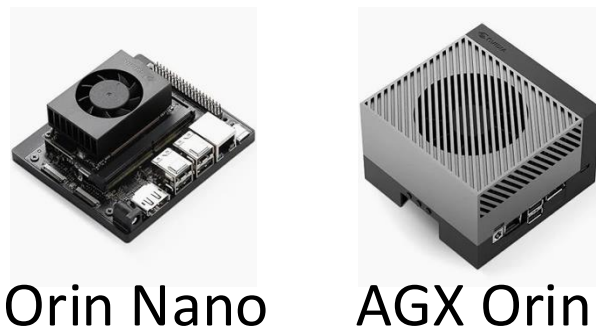
- Mechanism 2: A Closed-loop RL Controller



Runtime signals → joint frequency and decoding control.

Experiments and Results

- Setups:

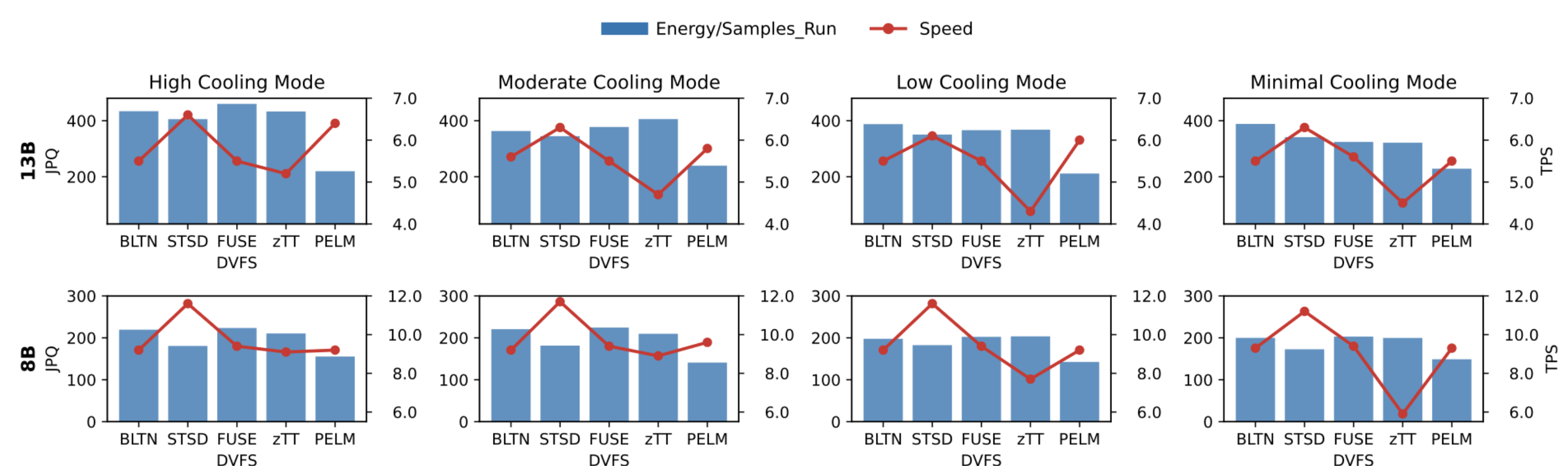


Orin Nano AGX Orin

DVFS Method	High	Moderate	Low	Minimum
BLTN	1.00	1.00	0.70	0.45
STSD	1.00	1.00	1.00	0.54
FUSE	1.00	1.00	0.70	0.43
zTT	1.00	1.00	1.00	1.00
PELM	1.00	1.00	1.00	1.00

Request Completion Rate Under Different Cooling Conditions

- PELM maintains **100%** completion rate.



Energy Consumption and Decoding Speed Comparison

- Up to **52.4%** energy reduction and **23.1%** speedup.
- Large gains appear under heavier workloads.

Takeaways

How hard to work

PELM controls CPU/GPU frequency through DVFS.

How much to work

PELM scales LLM decoding workload using variable-depth speculative decoding.

Why it matters

Joint hardware–decoding control reduces energy and thermal pressure for on-device LLM inference.

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